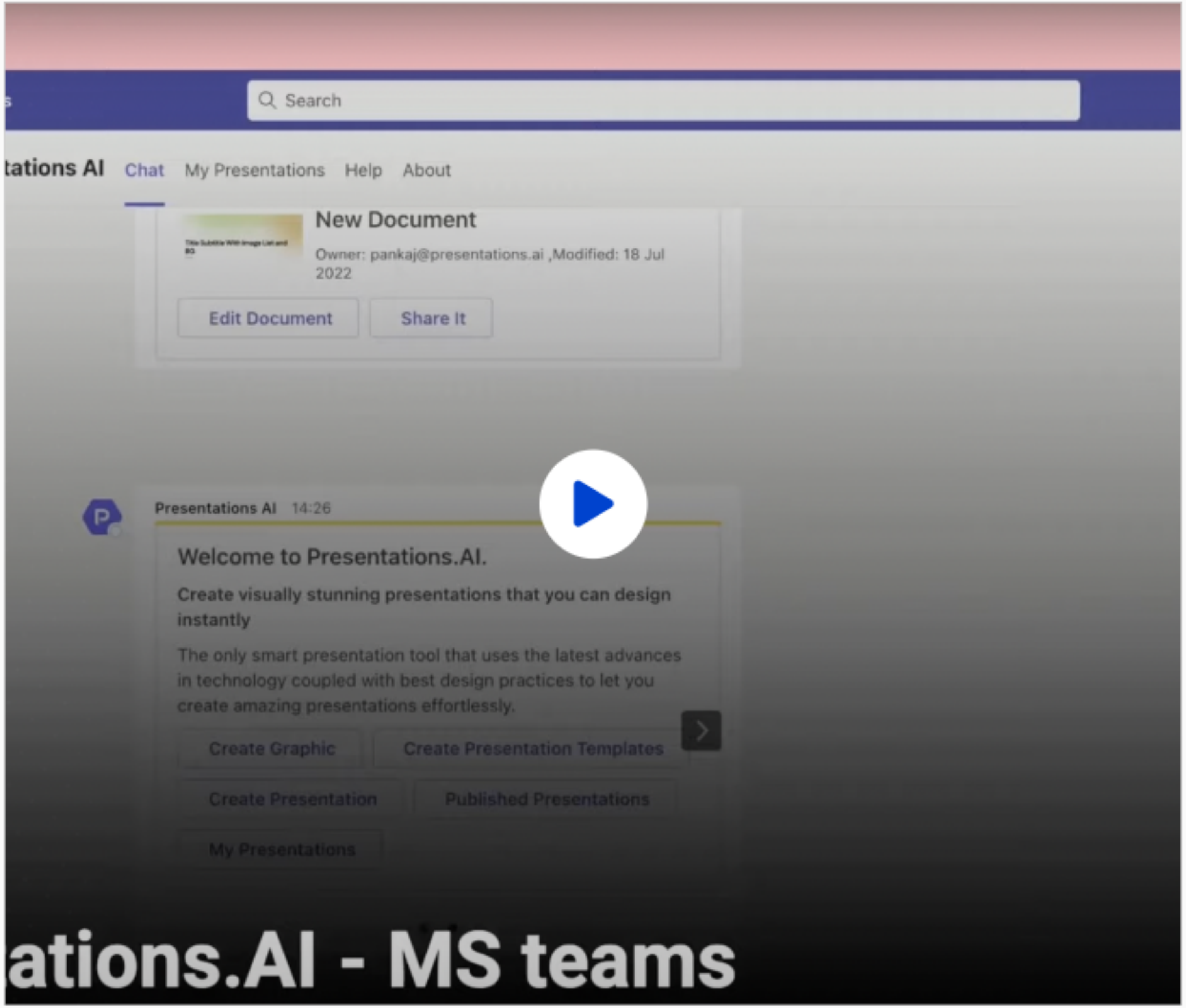
Fake News Detection Using BERT

Nikita Agarwal
Nitin Negi
Poornima
Prasannajeet



INTRODUCTION

This project explores the use of BERT-based transformer models to detect fake news in digital media. Fake news spreads rapidly online, influencing public opinion and societal trust. Traditional detection methods struggle with nuanced language and context, making advanced NLP techniques essential.

We fine-tune DistilBERT, a lightweight variant of BERT, on a balanced dataset of real and fake news articles. The model captures contextual semantics to classify news accurately, aiming to provide a scalable and efficient solution for misinformation detection.

Motivation

1. Information Age: Social media enables instant information sharing but also accelerates fake news dissemination, impacting politics, health, and social trust.
2. Need for Automation: Manual fact-checking is impractical at scale; intelligent systems using NLP and deep learning are essential for real-time fake news detection.
3. Rise of Transformers: BERT's bidirectional context understanding surpasses traditional ML models, offering improved accuracy in detecting nuanced misinformation.

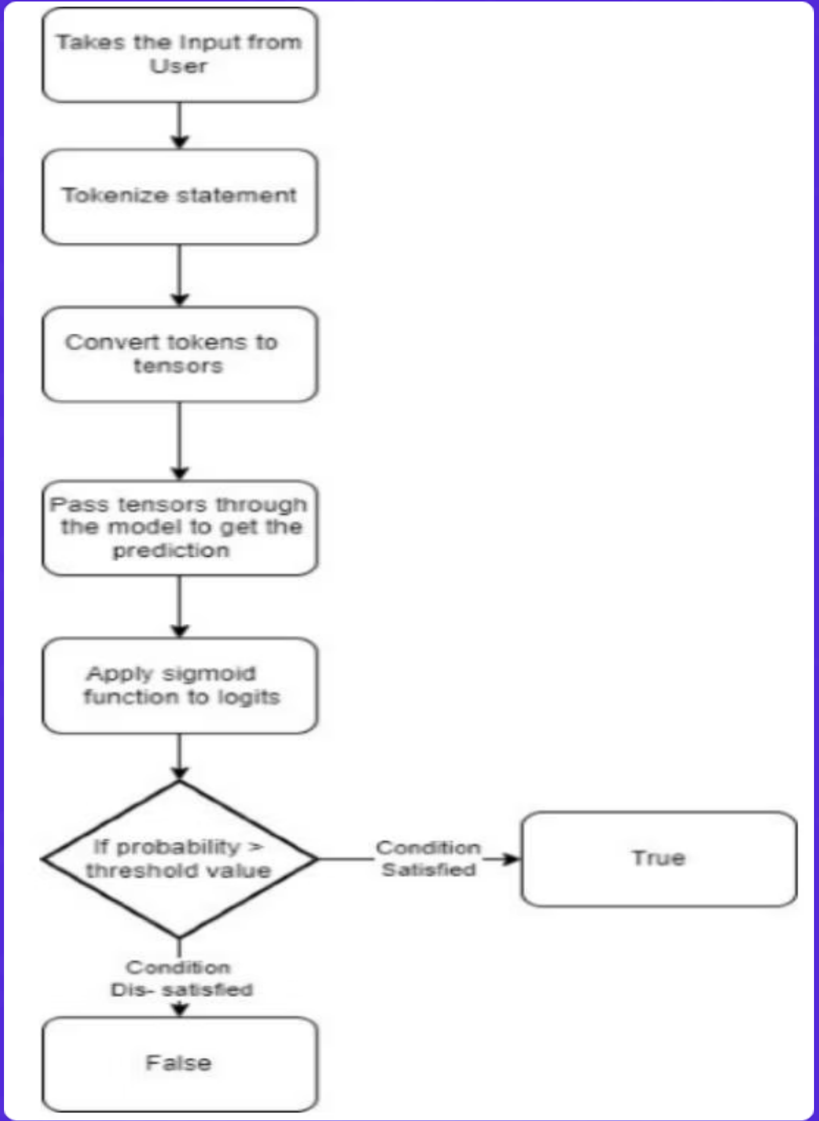
Objectives and Key Outcomes

- Model Development: Fine-tune DistilBERT to classify articles as real or fake with high accuracy.
- Performance Evaluation: Use metrics like accuracy, precision, recall, F1-score, and confusion matrices to validate model effectiveness.
- Deployment Readiness: Save model and tokenizer for future integration into real-time applications and research extensions.
- Research Contribution: Provide a scalable, interpretable foundation for combating misinformation using transformer-based NLP.

Literature Review: Evolution of Fake News Detection

- Traditional Machine Learning: Methods like Naive Bayes, SVM, and Random Forest used handcrafted features but lacked semantic understanding.
- Deep Learning Advances: RNNs, LSTMs, and CNNs improved feature extraction but struggled with long-range dependencies and context.
- Transformer Models: BERT and variants capture bidirectional context, outperforming earlier models in fake news classification tasks.
- DistilBERT: A smaller, faster BERT variant retaining most accuracy, enabling efficient deployment in resource-constrained settings.

Projective Description and Methodology



- Data Collection: Use Kaggle dataset with labeled fake and real news articles, balanced for binary classification.
- Preprocessing: Clean text, tokenize using DistilBERT tokenizer, pad/truncate sequences to fixed length.
- Model Fine-tuning: Train DistilBERT with binary cross-entropy loss and AdamW optimizer on tokenized data.
- Evaluation: Assess accuracy, precision, recall, F1-score, and visualize confusion matrix and ROC curves.

Code Implementation and Tools

Python 3.10 is used for data processing and model training.

PyTorch serves as the deep learning framework for DistilBERT.

The Transformers Library by Hugging Face provides pre-trained models and tokenizers.

Matplotlib and Seaborn are utilized for visualizing performance metrics and confusion matrices.

Data Analysis and Insights

- 😀 Dataset Overview: Balanced dataset with fake and real news articles, combined and shuffled for training.
- 😀 Average article length: words; Max length: words.
- 😀 Max token length set to for model input.
- 😀 Textual Patterns: Fake news often uses sensational keywords while real news focuses on official terms. Fake news shows stronger sentiment polarity.

Model Performance and Results

High Accuracy

Model achieved 96% accuracy with balanced precision and recall, indicating reliable classification.

Confusion Matrix Highlights

Low misclassification rates with 4 false positives and 5 false negatives.

ROC and Precision-Recall

AUC-ROC score of 0.97 demonstrates excellent separability between fake and real news classes.

AUC-ROC Score

0.97

AUC-ROC

Future Enhancements and Conclusion

Expand Language Support

Integrate multilingual models like mBERT for diverse languages.

Real-Time Integration

Connect with live news streams and social media APIs for continuous monitoring.

Explainable AI

Incorporate tools like LIME or SHAP for transparency in model decisions.

Deployment

Develop web/mobile apps and browser extensions for public access and verification.